comp10002/comp20005
linear search

Support material for Chapter 7 of
Programming, Problem Solving, and Abstraction with C,
by Alistair Moffat

Prepared by Alistair Moffat, ammoffat@unimelb.edu.au

Searching for: 19

no | maybe | yes!

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
17	23	41	25	22	52	17	19	21	24	14	37

i = 0

n = 12

```
i ← 0
while i < n and A[i] != x do
    i ← i + 1
if i ≥ n then
    return notfound
else
    return i
```

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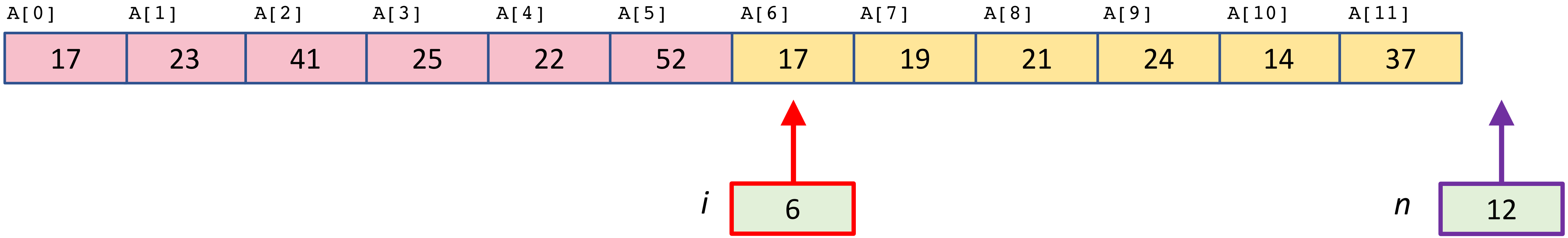

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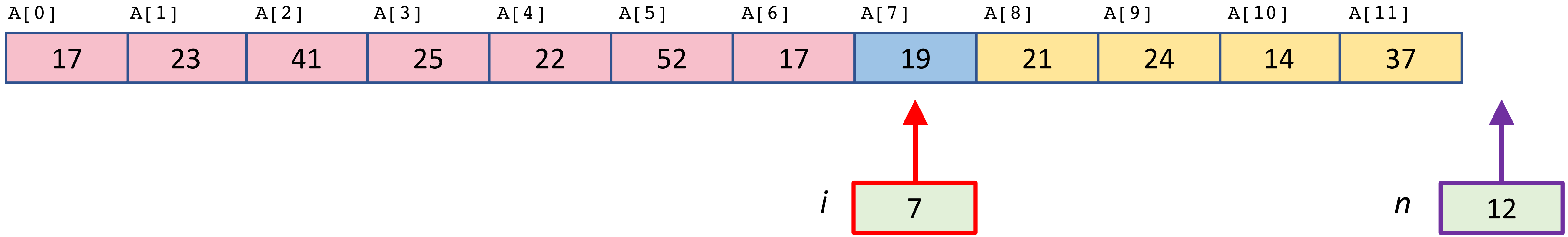

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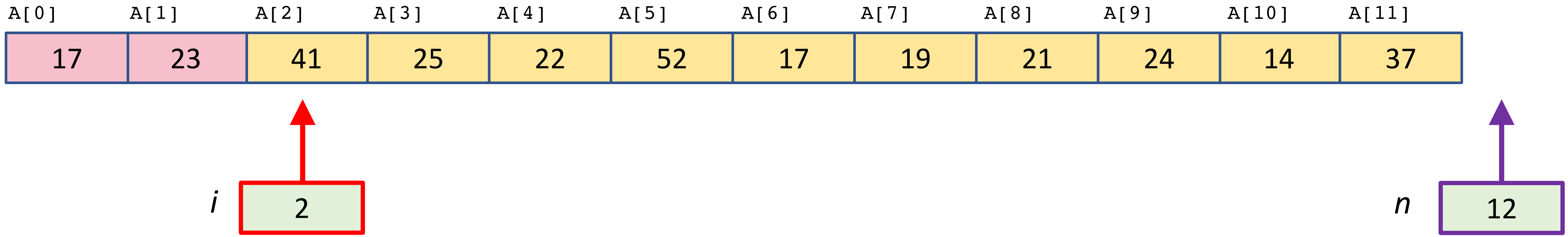

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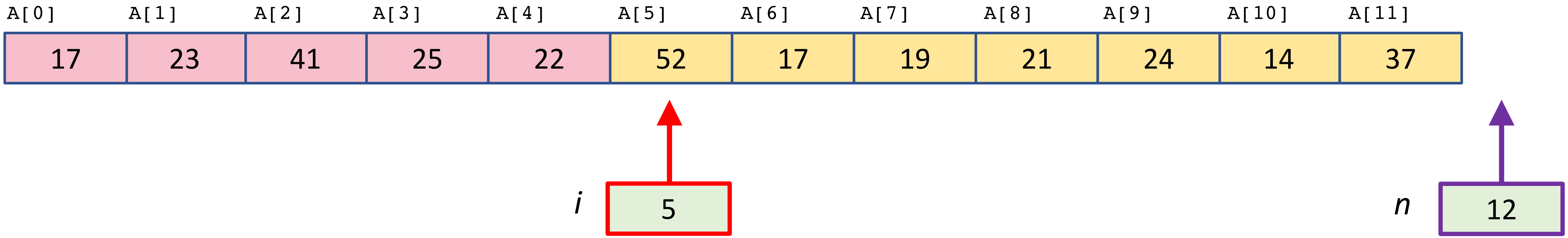

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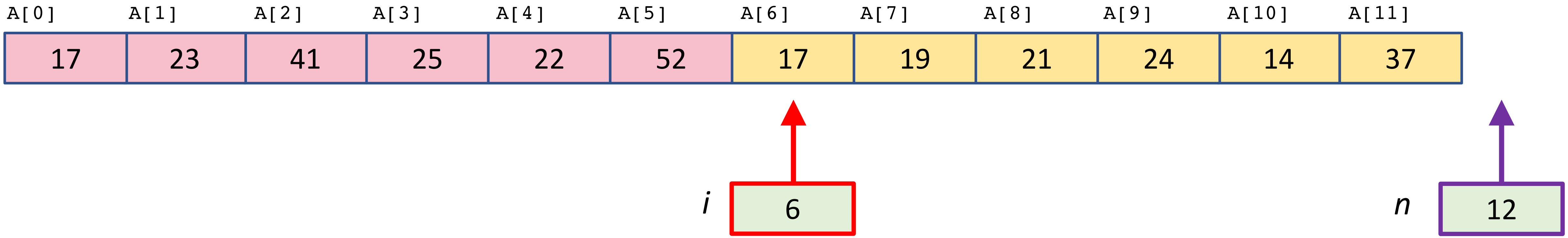

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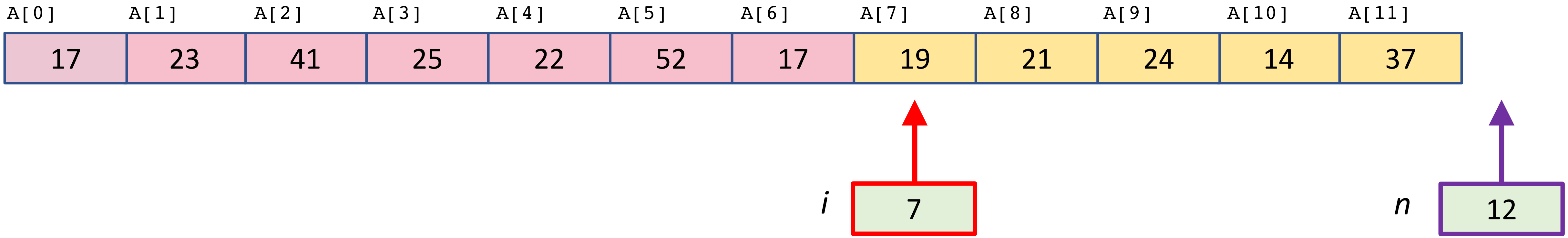

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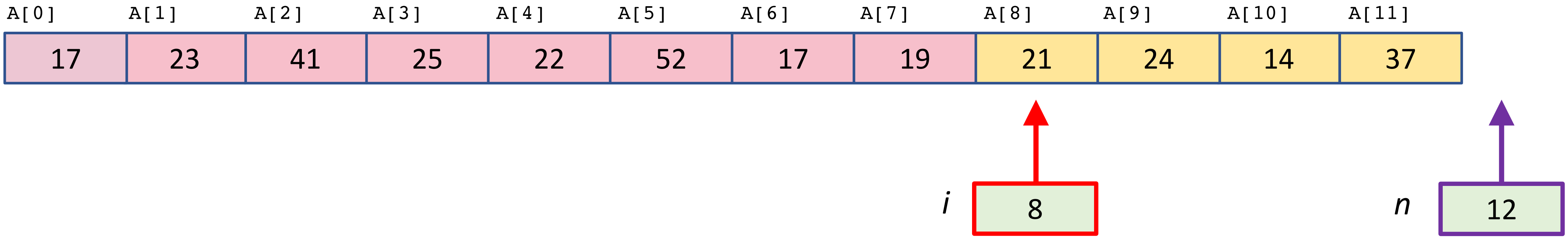

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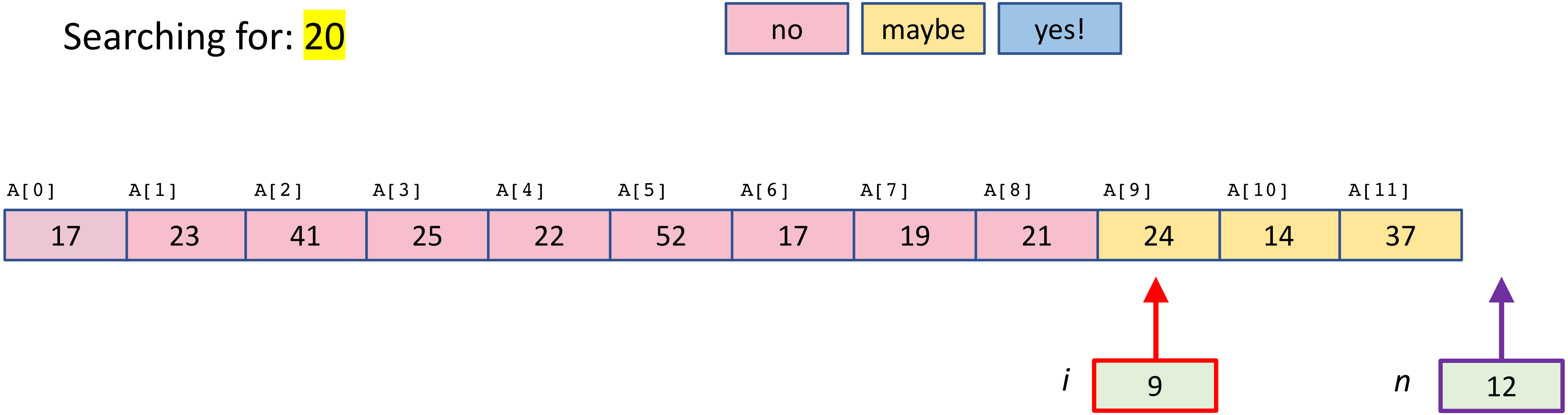

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while *i* < *n* **and** *A*[*i*] != *x* **do**
    i ← *i* + 1
if *i* ≥ *n* **then**
    return *notfound*
else
    return *i*